Résumé - Che Chen

(347) 465-7864 | cctom@umich.edu | Github: TomCC7

Education

University of Michigan *Sep. 2021 - May 2023 (expected)*

B.S.E in Computer Engineering, GPA: 3.97/4.00

Shanghai Jiaotong University - UM-SJTU Joint Institute *Sep. 2019 - Aug. 2023 (expected)*

B.S.E in Electrical and Computer Engineering, GPA: 3.26/4.00

Experiences

Robust Controller and System Identification On Robotic Arms *May 2022 - Present*

Research Assistant at ROAHM LAB

- Performed System Identification on Kinova Gen3 to measure the joint friction, damping, and transmission inertia
- Developed a robot control framework that provides both realtime performance and high-level abstraction
- Implemented a robust controller based on interval arithmetic and verified the tracking performance

Internship at Petoi LLC *May 2022 - Aug. 2022*

- Developed a ROS Interface to control robot powered by OpenCat
- Applied a human pose detection model to map human postures to robot cat [demo]

Realtime and Virtual Driving Simulator [code] *Feb. 2022 - Present*

Multidisciplinary Design Program guided by Prof. Paul Green

- Aims to create a realistic and handy simulation environment for non-programmer researchers
- Developed Joystick Driver on Linux for Carla Simulator
- Developed Wizard program to provide low-cost self-driving car simulation
- Leading software development of the backend team

Localization System Based on UWB (Ultra-Wide Band) Anchors *Feb. 2021 - July 2021*

SJTU Innovative Practical Project (IPP) advised by Prof. Mian Li

- Setup a localization system using UWB anchors
- Performed localization task using data from UWB anchors and IMU
- Used TCP protocol to send location data to a server and display it in Unity

Course Projects

Neural Radiance Field (NeRF) [code] *Apr. 2022*

Deep Learning for Computer Vision by Prof. Justin Johnson

- Learned how to setup PyTorch ML environment from scratch
- Learned key ideas on NeRF such as Volume Rendering methods

Table Tennis Ball Collector for Use in Multi-ball Training [code] *Feb. 2022 - Apr. 2022*

Introduction to Embedded System Design by Prof. Robert Dick

- Used UART protocol to communicate between raspberry pi and an STM32 controller board
- Implemented a CV algorithm to estimate the 3D position of ping pong balls using a single RGB camera
- Designed a simple but efficient ping pong ball collecter using 3D printing and rubber bands

Robot Localization in PyBullet [code] *Dec. 2021*

Introduction to Algorithmic Robotics by Prof. Dmitry Berenson

- Implemented a path planning algorithm using A* in Pybullet simulation
- Implemented a Kalman Filter and Particle Filter, and analyzed their performance in terms of accuracy

Skills

Programming C++, Python, MATLAB, Julia, HTML/CSS, Bash, LaTeX
Software ROS (1&2), Docker, Git, Emacs, Linux, Solidworks
Knowledge Robot Kinematics & Dynamics, Control System, Robot State Estimation, Computer Vision, Embedded System